NET1014 NETWORKING PRINCIPLES GROUP ASSIGNMENT REPORT

Prepared For:
Dr Maisarah Mansor

Group Report By:

Tan Zheng Lin – 23038482

Chen YiFei – 24101545

Radin Mikael Bin Radin Izuanudin - 25011859

Phan Jin Hao – 24008864

Muhammad Hussain - 23115942

SUNWAY UNIVERSITY

PART 1: COMPREHENSIVE ANALYSIS OF OSI PROTOCOL STACK

What is an OSI Model?

An Open Systems Interconnection (OSI) model is a framework that allows communication systems to communicate with each other using standard protocols (such as IP addresses). Created by the International Organization for Standardization, the OSI model is seen as a universal language within the field of networking. The model is split into seven layers, each that can communicate with their neighboring layers and handle a specific job.

Layer 1: Physical Layer

The physical layer is the first layer of the OSI model, as it lays as the foundation of all network communications. The physical layer is known as the foundation layer because a physical connection needs to be established before any communication can happen, it also handles available resources within communication which furthermore defines the relation between physical devices and transmission media (Das, 2018). During transmission, the physical layer defines the type of signal after converting, which can be electrical or optical signals (Jasud, 2017). There are different types of connections in the physical layer, such as wired (Ethernet and fiber-optic cables) and wireless (Wi-Fi and Bluetooth). The physical layer commonly uses the Ethernet protocol as it can be used for local area networking between devices as it can come in different configurations that make it scalable in the real world.

Layer 2: Data Link Layer

The data link layer is the second layer of the OSI model. The main purpose of the data link layer is to allow communication between devices that are within the same network to provide reliable methods to transfer data between nodes (Chen et al., 2010). The data link layer is split between two sub-layers, those being logical link control (LLC) as the upper sub-layer and media access control (MAC) as the bottom sub-layer. The LLC sub-layer's main function is to allow the bridging between the networking layer and the MAC sub-layer, this is so that protocols are allowed to share the same link to ensure a smooth and reliable data flow. The MAC sub-layer's main function is to manage the network physical connection at the data link layer, while also providing physical locations to devices in the form of a ‘MAC address’ for data deliveries. The protocols used in the data link layer are Ethernet cables and Wi-Fi. There are also MAC protocols such as CSMA/CD and CSMA/CA that control access to shared media.

Layer 3: Network Layer

The network layer is the third layer of the OSI model. The main purpose of the network layer is to deliver data between devices that are of different networks. The network layer ensures connections between devices using IPs to provide a logical location and provides data delivery through routing so that data is able to transfer itself to other networks. Other than IPv4 or IPv6, the layer also consists of protocols such as ICMP, ARP, and routing protocols such as OSPF. The OSPF protocol is a part of dynamic routing, where routers can exchange information about the view of the network topology (GeeksforGeeks, 2025). The OSPF protocol uses link-state algorithms to find the best path for data delivery.

Layer 4: Transport Layer

The transport layer is the fourth layer of the OSI model. It serves as the middle ground between the network layer and the application layer, where it accepts data that comes from the session layer and splits the data to pass into the network layer (Mishra, 2017). Furthermore, they break application messages into segments that pass into the network layer while also reassembling segments into messages that could pass into the session layer (Ross & Kurose, 2013). The protocols used within the transport layer are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP). The TCP is more reliable than UDP, as it is able to provide reliable, in-order delivery between processes while UDP is simply faster and more efficient than TCP as it does basic error-checking (BasuMallick, 2025).

Layer 5: Session Layer

The session layer is the fifth. Session layer is a key for the networking communication because it limits unnecessary processing power usage and conserves bandwidth, which results in more effective connections. This layer is responsible for opening, closing, and managing communication sessions between local and remote hosts. By making sure the connections are limited only when it is necessary, the Session layer allows data to be transmitted efficiently then closes them once communication is complete, helping to preserve network resources. Session Layer functionality is commonly associated with a few protocols. Some of the protocols are Point-to-Point Tunneling Protocol (PPTP), Session Description Protocol (SDP), and Remote Procedure Call Protocol (RPC) (Raza, n.d.).

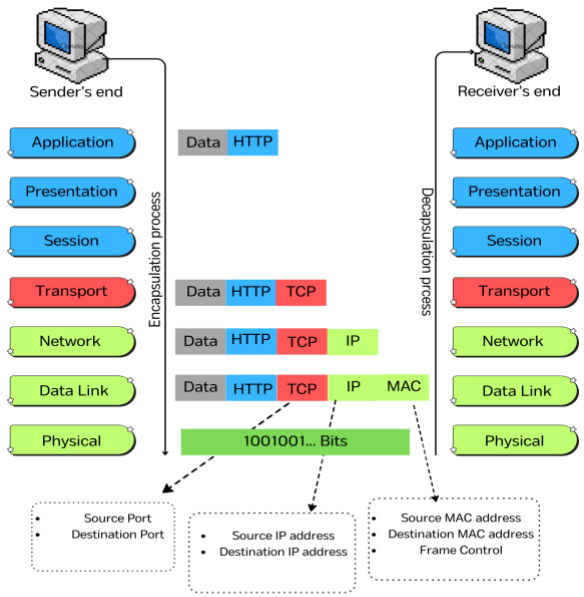
Layer 6: Presentation Layer

The presentation Layer is the sixth layer of the OSI model. It acts as a translator for the network, ensuring data that is exchanged from the application layer and the network, is in a readable format where both systems are able to interpret. At the sender's end, the data are encrypted and formatted before the transmission process happens while at the recipient's end, the data are decrypted and translated for the use of the Application Layer. By facilitating interoperability across multiple systems and maximizing bandwidth utilization, these procedures facilitate effective and secure end-to-end communication. Transport Layer Security (TLS) and Secure Socket Layer (SSL) are used for encryption. They are two real-world protocols that work with Presentation Layer functionality.

Layer 7: Application Layer

The seventh layer of the OSI model is the application Layer. It provides network services to end users. Some of the essential services are email communication, remote access, file transfer, and directory services, allowing users to send messages, transfer files, and access network resources. This layer defines interfaces and protocols for data exchange, allowing user-facing apps to communicate over a network. Application layer interacts with the presentation layer to send and receive information in a usable form for the user. Some common real-life protocols operating at this layer are HyperText Transfer Protocol (HTTP) for web communication and Domain Name System (DNS) for translating domain names to IP addresses.

Diagrams showing encapsulation and data flow between layers



Advantages: Foundational Clarity and Standardization

The main advantage of the OSI model is its role in standardization and education. It was designed for break the manufacturer's lock, and through establishing a common standard for data networking protocols to realize allow communication between disparate networking devices across all over the word (*OSI Model Advantages and Basic Purpose Explained*, 2025). Its layered architecture provides significant huge operational benefits. By dividing the process of complex data exchanging to seven different parts, it simplifies understanding, management, and troubleshooting for learners and administrators alike.

Disadvantages: Structural and Practical Limitations

However, the model's theoretical purity is also the source of its main criticisms. Which leads to complexity and inefficiency. It is worth noting that it defines similar service such as error control, superfluous across multiple layers, which violates the principle of layer independence. Besides, its strict hierarchical design cannot allow layers to work simultaneously, which negatively impacts performance (*OSI Model Advantages and Basic Purpose Explained*, 2025). In practical, this model always defined too detailed; it defines every protocol and function meticulously, that reduces the direct implementation guide for network engineers.

PART 2: NETWORK DESIGN

Required Equipment and Justifications:

- 2 Cisco 2911 Routers both with a HWIC-2T card
- 4 Cisco 2960-24TT switches
- Required cables such as serial, copper straight through etc.
- Server Racks
- Pegboards
- Equipment Shelves
- Tables
- Computers

Cisco 2911 routers are used as they provide sufficient ports for connecting to switches, as well as continued support from Cisco in terms of software. When equipped with the HWIC-2T card, the routers are able to support a serial WAN connection between the HUMAC network and SET's New Building network, making them suitable for inter-building communication.

Cisco 2960-24TT switches are used as they are easy to configure and reliable in handling network traffic across both locations. They also provide enough ports to support all laboratory hosts and are space-efficient, as each switch occupies only one slot in a standard server rack.

Serial cables are used to establish the WAN connection between the two routers, while copper straight-through Ethernet cables are used for router-to-switch and PC-to-switch connections. These cable types are standard for local and wide area network setups and ensure stable data transmission.

Server racks, equipment shelves, and pegboards are included to allow secure mounting of networking devices and to support proper cable management. Tables are used to support the placement of computers and other end devices within each laboratory. Computers serve as end hosts for users in each lab and are required to test and validate network connectivity.

Estimated Total Cost:

- 2 × Cisco 2911 routers: $1,770

- 4 × Cisco Catalyst 2960-24TT switches: $6,000

- Network cables (serial and copper straight-through): approximately $200

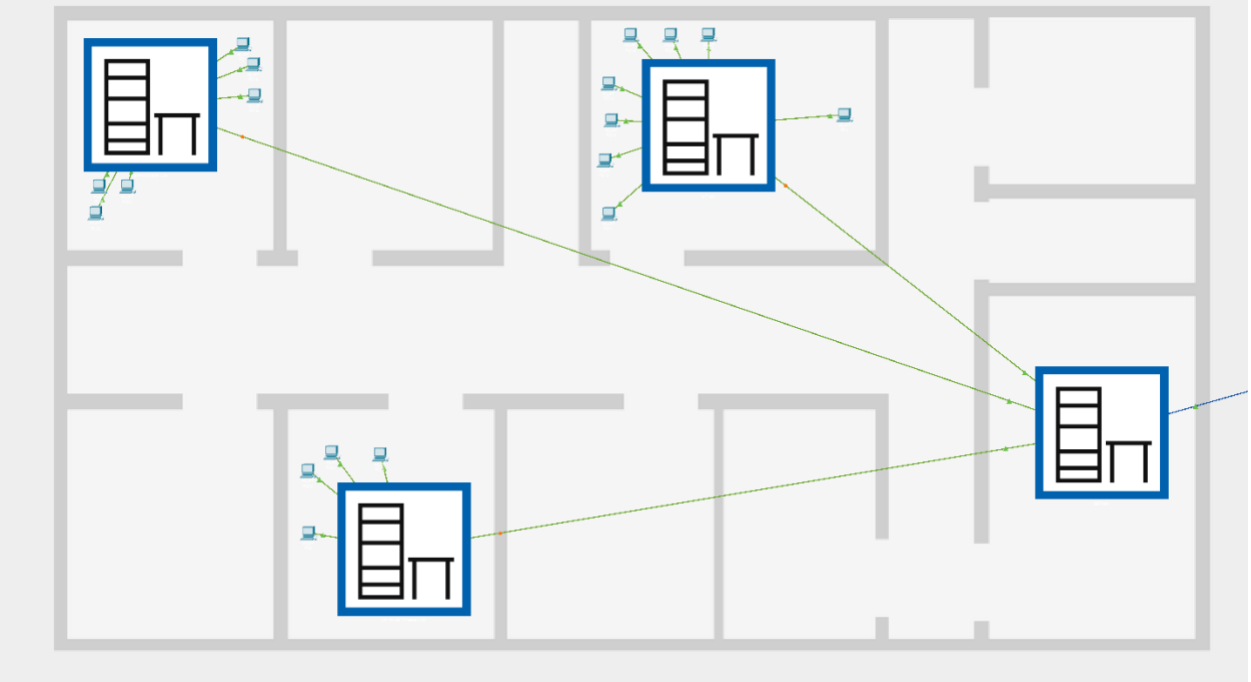
- Server racks, equipment shelves, pegboards, and tables are assumed to be existing university infrastructure and are therefore not included in the total cost. Computers are also excluded from the costing as they are provided separately for laboratory use.

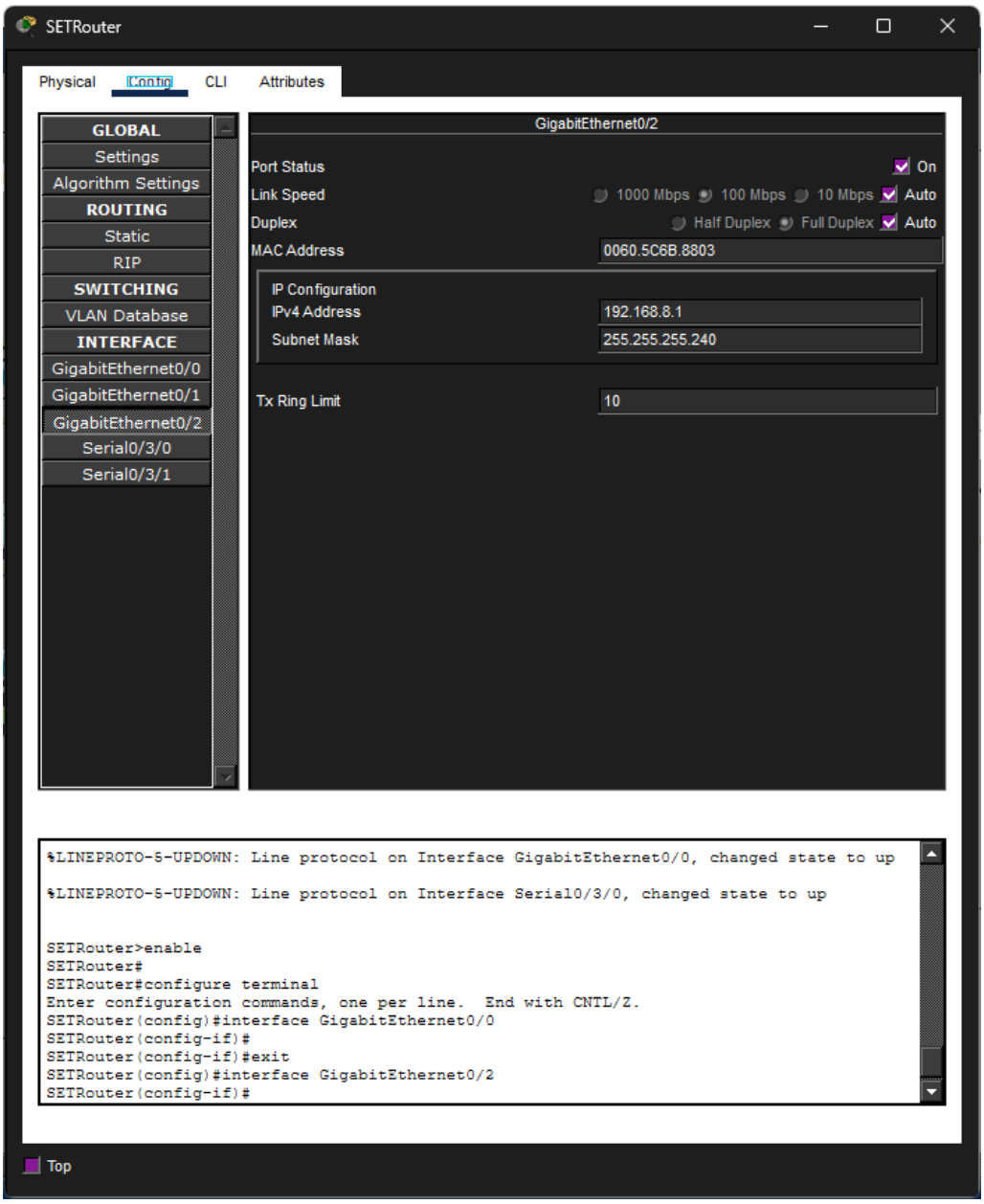
- The estimated total cost for the networking equipment required in this design is approximately **$7,970**

Physical and Logical Topology:

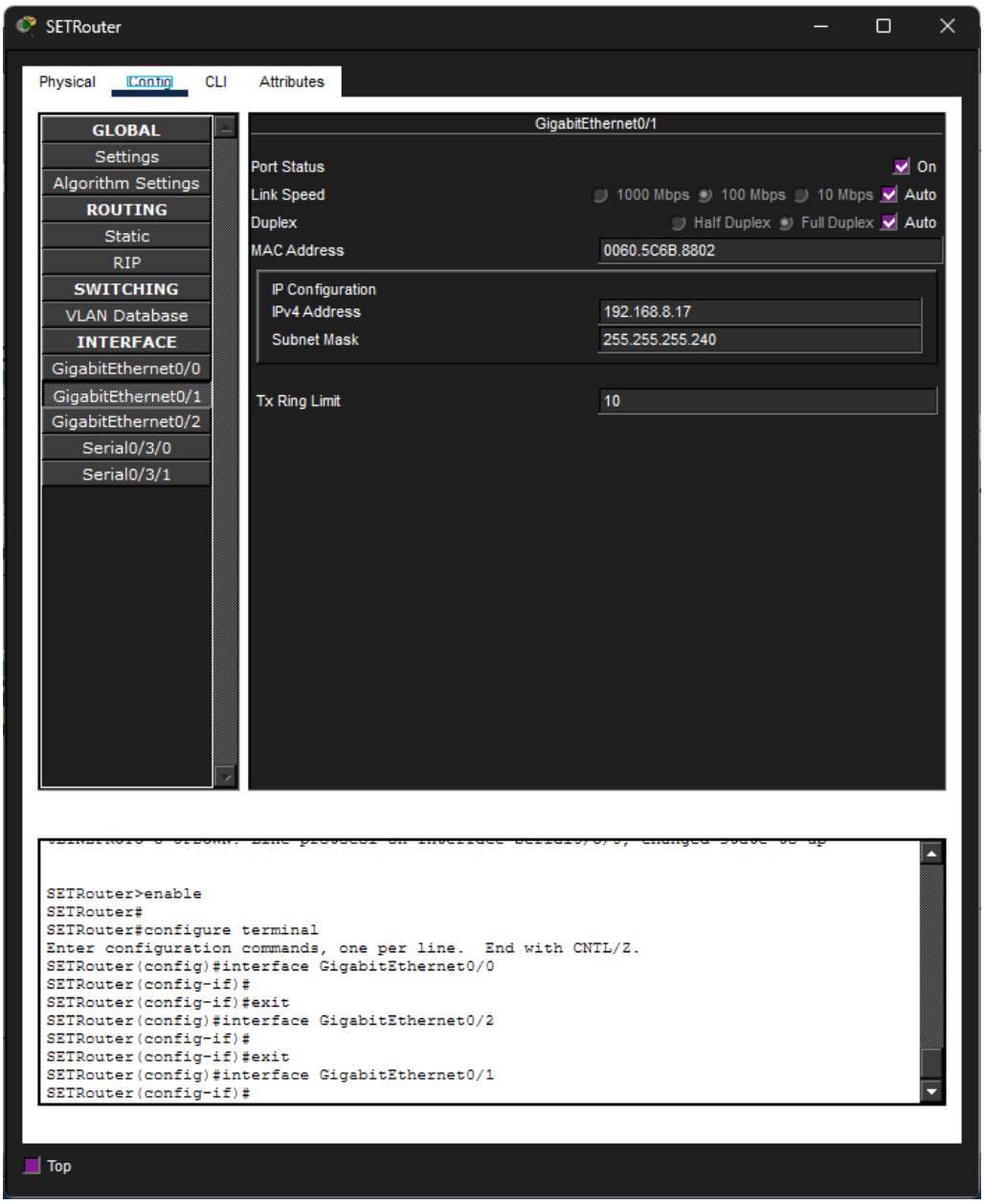
SET New Building

NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
IoT Lab	9	14	5	192.168.8.0	/28	255.255.255.240	192.168.8.1 - 192.168.8.14	192.168.8.15
Communication Lab	7	14	7	192.168.8.16	/28	255.255.255.240	192.168.8.17 - 192.168.8.30	192.168.8.31
Advanced Wireless Lab	5	6	1	192.168.8.32	/29	255.255.255.248	192.168.8.33 - 192.168.8.38	192.168.8.39

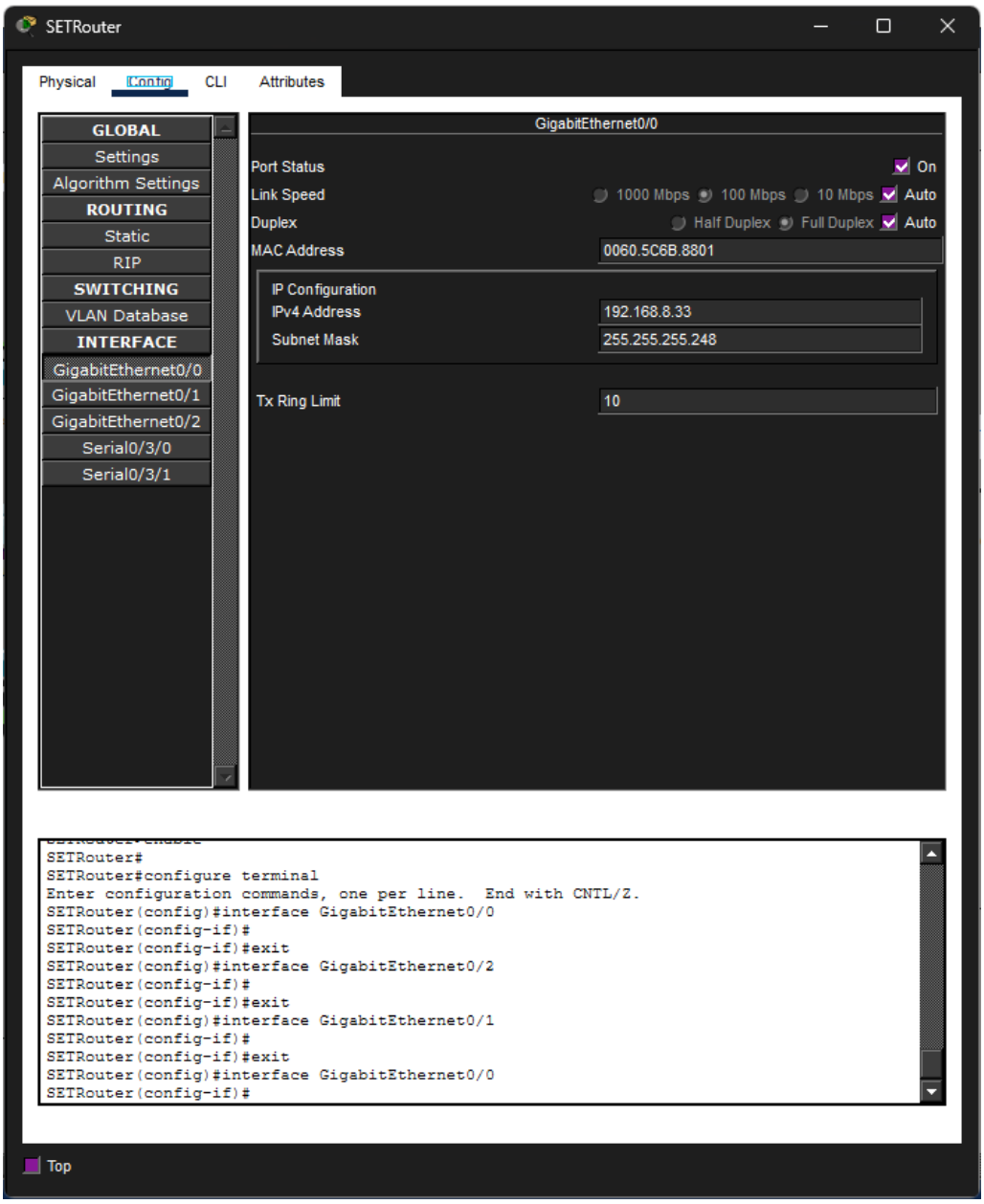




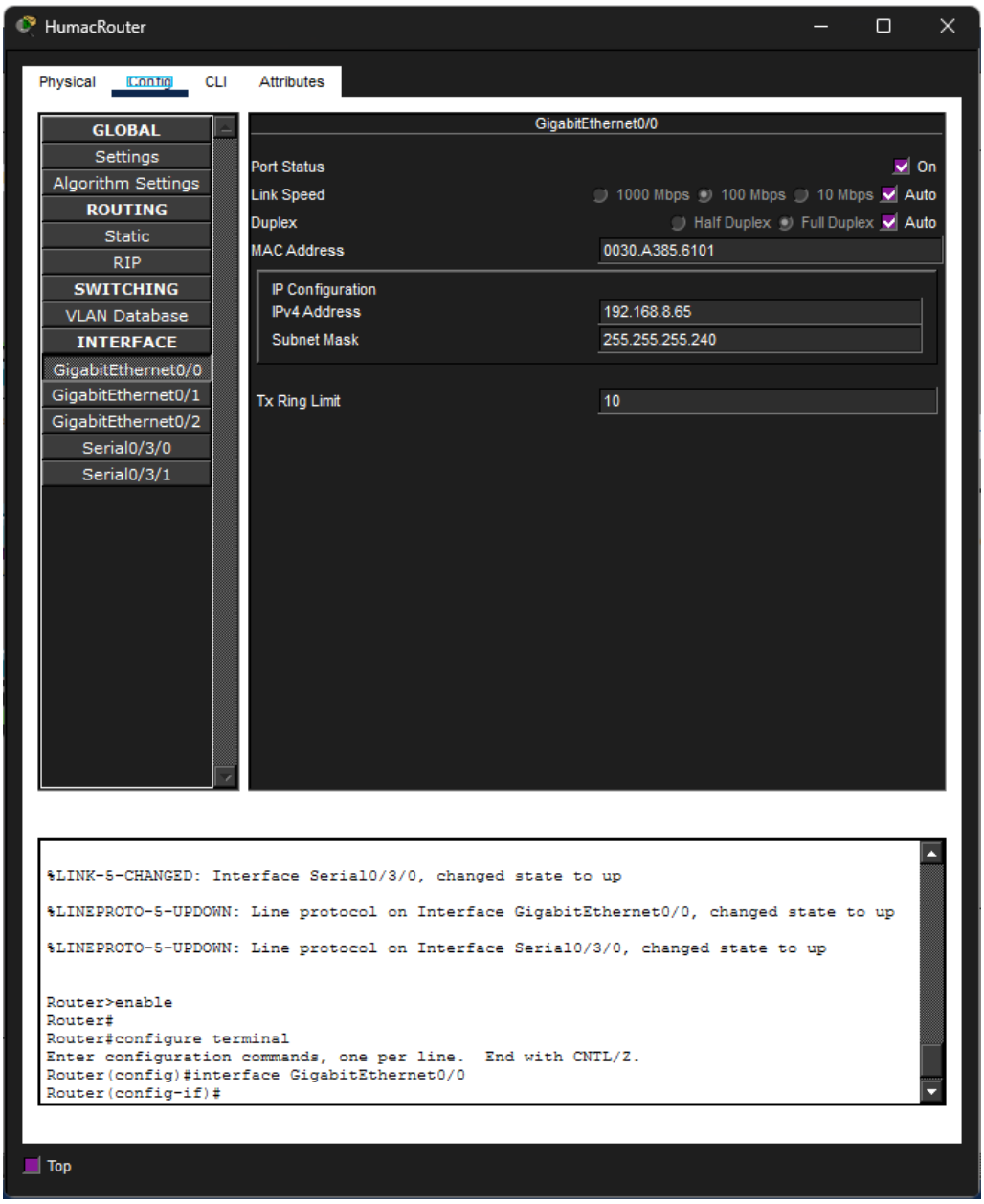
IP Address and configuration of IoT Lab Switch



IP Address and configuration of Communication Lab Switch



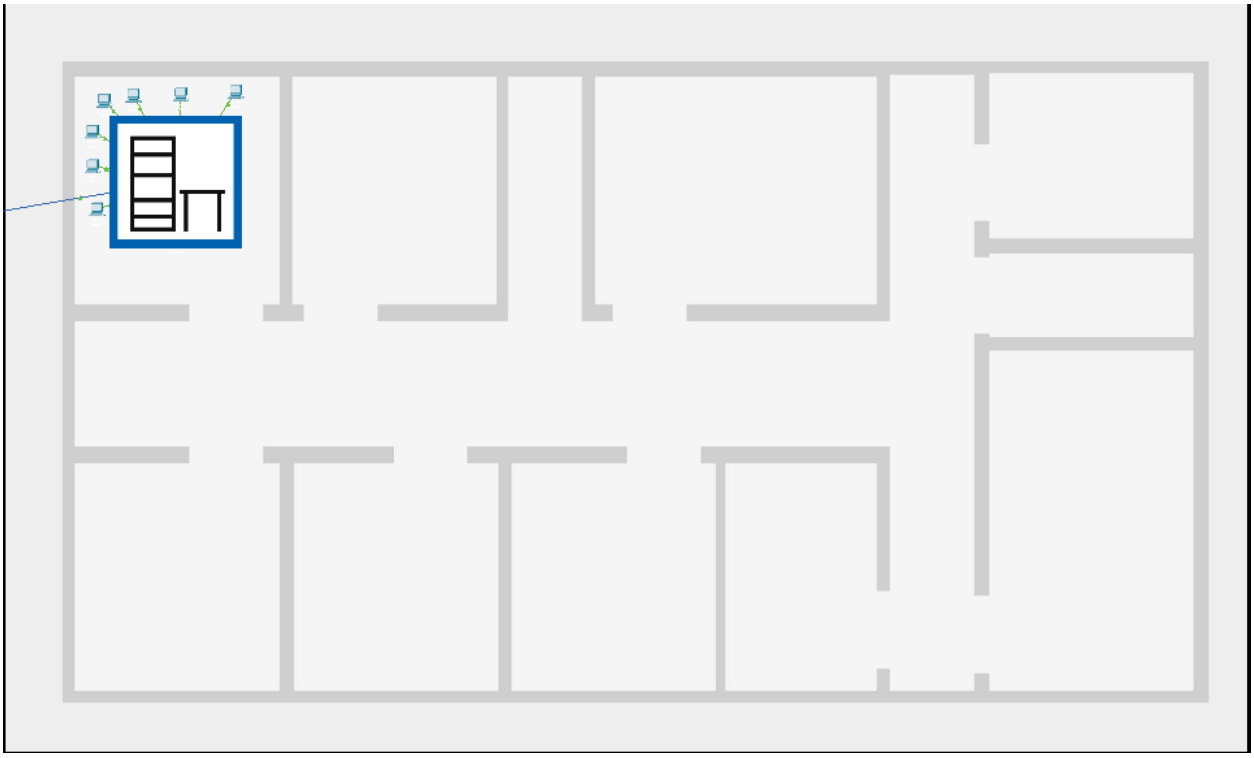
IP Address and configuration of Advanced Wireless Lab



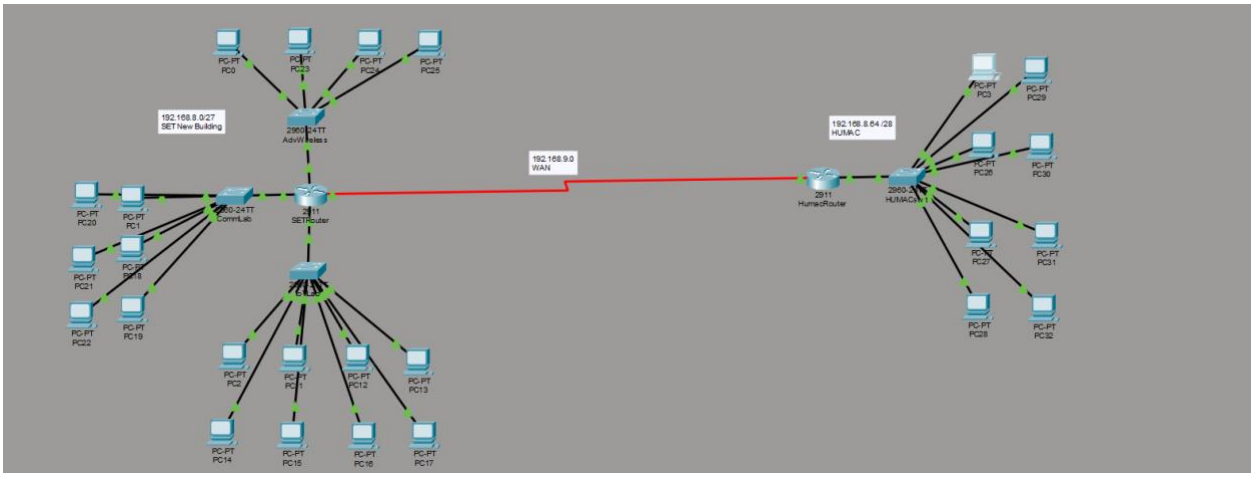
IP address and configuration of HUMAC Labs Switch

HUMAC Lab

3rd Floor	9	14	5	192.168.8.64	/28	255.255.255.240	192.168.8.65 - 192.168.8.78	192.168.8.79



PART 3: NETWORK IMPLEMENTATION



Configuration Steps:

- Routers, Switches and Hosts on the SET New Building were first set up to have their own IP addresses starting from 192.168.8.1
- Routers, Switches and hosts on the HUMAC side were then set up to have their own IP addresses starting from 192.168.8.65
- Routing has been done through 192.168.9.0 and are configured for both routers
- Ping tests and traceroutes have been done to ensure that all hosts are interconnected. First ping usually fails as it is used for ARP but second ping is successful for all devices.

PART 5: INDIVIDUAL REFLECTION AND ANALYSIS

Role and Contributions of Each Member:

- Tan Zheng Lin: Cisco Packet Tracer Model, Physical and Logical Topologies
- Radin Mikael: Explanation of Layer 5-7 of OSI model, created the OSI Model labelled diagram showing encapsulation and data flow between layers.
- Phan Jin Hao: Explanation of Layer 1-4 of OSI model, Completing group reflection
- Chen Yifei: Critically evaluate the advantages and limitations of the OSI model in today's networking environment
- Muhammad Hussain: VLSM calculations and IP address allocation for the network design, as well as contributions to equipment justification and cost estimation

This final assessment has helped us understand the fundamental concepts of networking in Cisco Packet Tracer. During the assessment, each member was given different tasks to handle different parts of the networking design, while also were given deadlines to complete. However, there are several issues that had stalled the progress of the design, such as pinging issues where pinging another device cannot be provided, or routing issues where routers were misconfigured, and troubleshooting issues where devices were malfunctioning during the process. As a solution, we have found that there are devices where their IP address was completely misconfigured, or there are missing digits within the subnet mask. Throughout this assessment, we have learnt how to create subnets based on the hosts' requirements, assigning IP addresses, and configuring subnet masks and default gateways to allow communication between end-devices through the creation of addressing tables to reference.

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